CMSC203 Assignment 6 Design

Class: CMSC203 CRN 30295/32324
Program: Assignment 6
Instructor: Professor Grinberg
Summary of Description: Circus!
Due Date: 04/27/2026
Integrity Pledge: I pledge that I have completed the programming assignment independently. I have not copied the code from a student or any source.

Part1:

Pseudo Code:

Circus Class, Add Animal

```
ADD the animal to the animals list
```

Circus Class, display AllAnimals

```
FOR each animal in the animals list
    DISPLAY animal details
```

Circus Class, addPerson

```
ADD the person to the persons list
```

Circus Class, displayAllPersons

```
FOR each person in the persons list
    DISPLAY person details
```

Circus Class, addBuilding

```
ADD the building to the buildings list
```

Circus Class displayAllBuildings

```
FOR each building in the buildings list
    DISPLAY building details
```

Circus Class, sortAnimalsByAge (Selection sort)

```
FOR i from 0 to size of animals list minus 2
    SET indexOfSmallest to i
    FOR j from i+1 to size of animals list minus 1
```

```
        IF age of animal at position j is less than age of animal at
indexOfSmallest THEN
            SET indexOfSmallest to j
    IF indexOfSmallest is not equal to i THEN
        SWAP animal at position i with animal at position
indexOfSmallest
```

Circus Class, sortAnimalsByName (Selection sort)

```
FOR i from 0 to size of animals list minus 2
    SET indexOfSmallest to i
    FOR j from i+1 to size of animals list minus 1
        IF name of animal at position j comes before name of animal
at indexOfSmallest alphabetically THEN
            SET indexOfSmallest to j
    IF indexOfSmallest is not equal to i THEN
        SWAP animal at position i with animal at position
indexOfSmallest
```

Circus Class, searchAnimalByName

```
FOR each animal in the animals list
    IF animal name equals the search name THEN
        RETURN the animal
RETURN not found (null)
```

Ticket price calculation

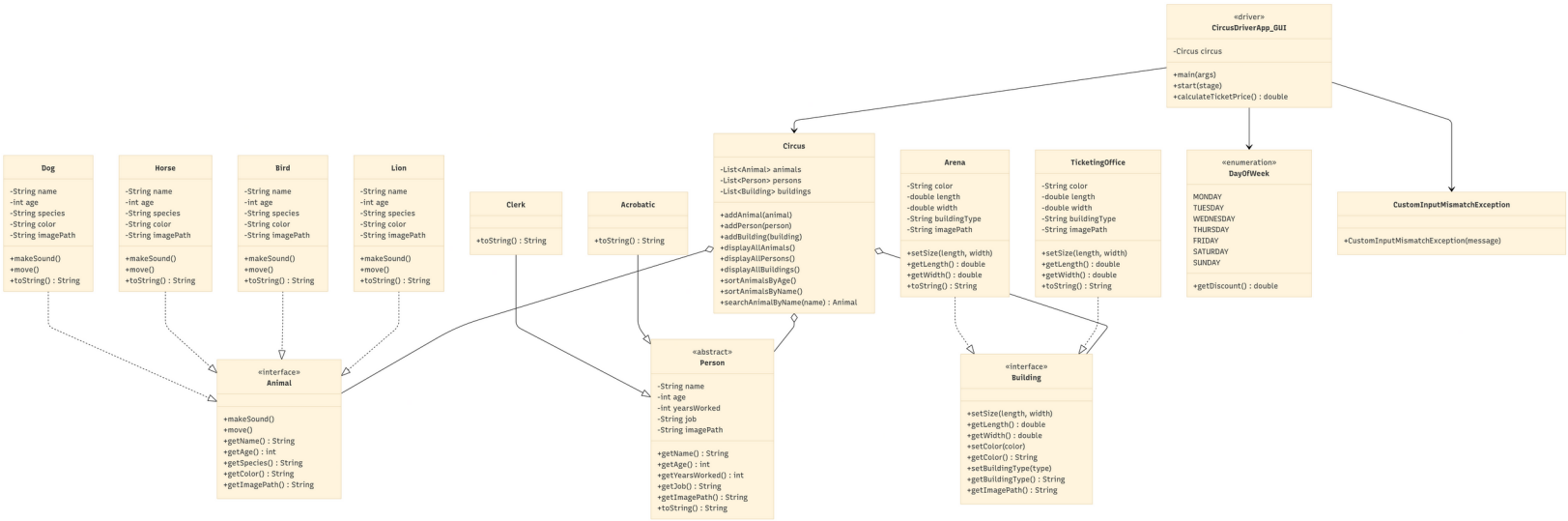
```
GET day discount from day of week selection
GET customer discount from customer type selection
GET seat multiplier based on seat level
    IF seat is Lower Level THEN multiplier is 1.0
    IF seat is T-Level THEN multiplier is 1.5
    IF seat is Upper Level THEN multiplier is 2.0
COMPUTE ticket price as base price times seat multiplier
COMPUTE ticket price minus day discount
COMPUTE ticket price minus customer discount
RETURN final ticket price
```

Bird / Lion / Arena / TicketingOffice, getters and setter template / pattern

```
GET [field name]
    RETURN [field name]

SET [field name]
    SET [field name] to the given value
```

Clerk Constructor / Acrobatics constructor

```
CALL the Person constructor with name, age, yearsWorked, job,
imagePath
```

Class UML Diagram